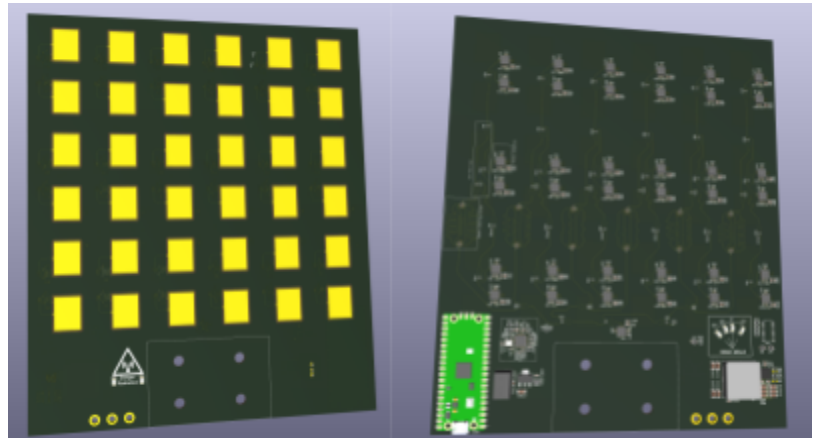


Steerable Microwave Beam Phased Array

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Document revision: 1.3

Status: In design



1. Project Overview

This project explores the design and construction of a steerable microwave phased array capable of delivering a tightly focused beam of microwave energy toward a target. This project will be my submission for the [OSHWLab Stars 2026](#) PCB design competition.

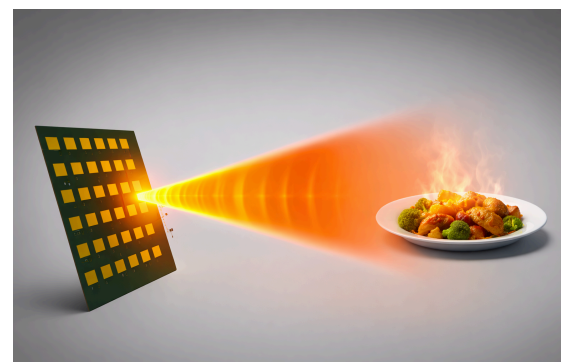
1.1. Purpose

The purpose of this project is to develop and **improve my technical skills** in areas that excite me, all while building a cool project that will **show flashy, mind-bending results**. The system serves as a science demonstration platform that reveals the power and elegance of microwave circuit design, phased arrays, and high-frequency PCB engineering. The core concept is to electronically steer RF energy without any moving mechanical parts, using phase control across an array of patch antennas.

1.2. Key Results

A successful project will have the following results:

- Delivers focused microwave energy across a distance
 - heats up food, illuminates LEDs across a room
- Focuses energy beam as tightly as possible
 - $\sim 17^\circ$ beam waist or 0.5m diameter at 2m
- Implementation of phased-array beamforming
 - Steers over a range of 75° in 1 axis
 - 4 programmable angles
- Extra things to show off (stretch goals)
 - Plasma generation inside noble gases
 - Recover radiated energy using a receiver antenna array (wireless drone charger?)

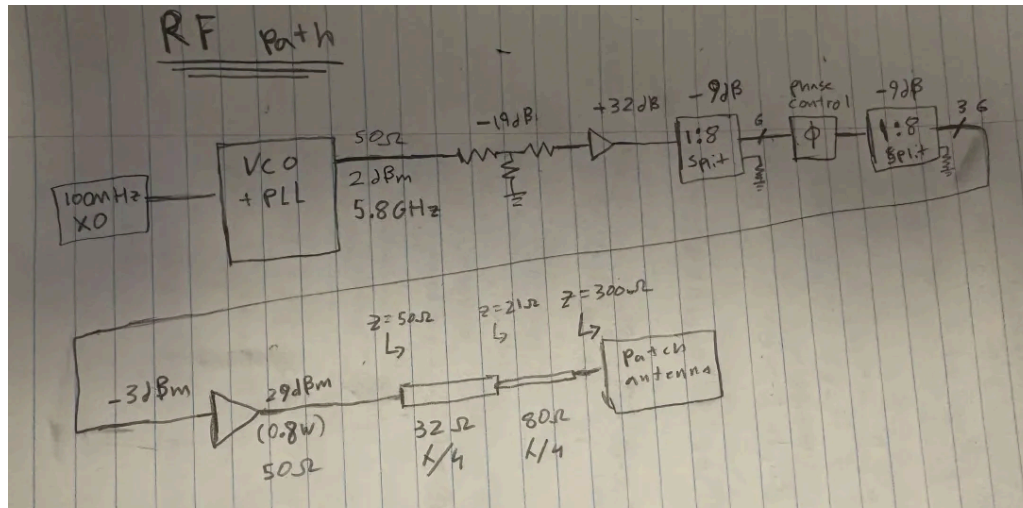


2. System Architecture

The beam is formed using a phased array of microstrip patch antennas. Each antenna element is driven with controlled phase offsets, allowing constructive interference in a desired direction and destructive interference elsewhere.

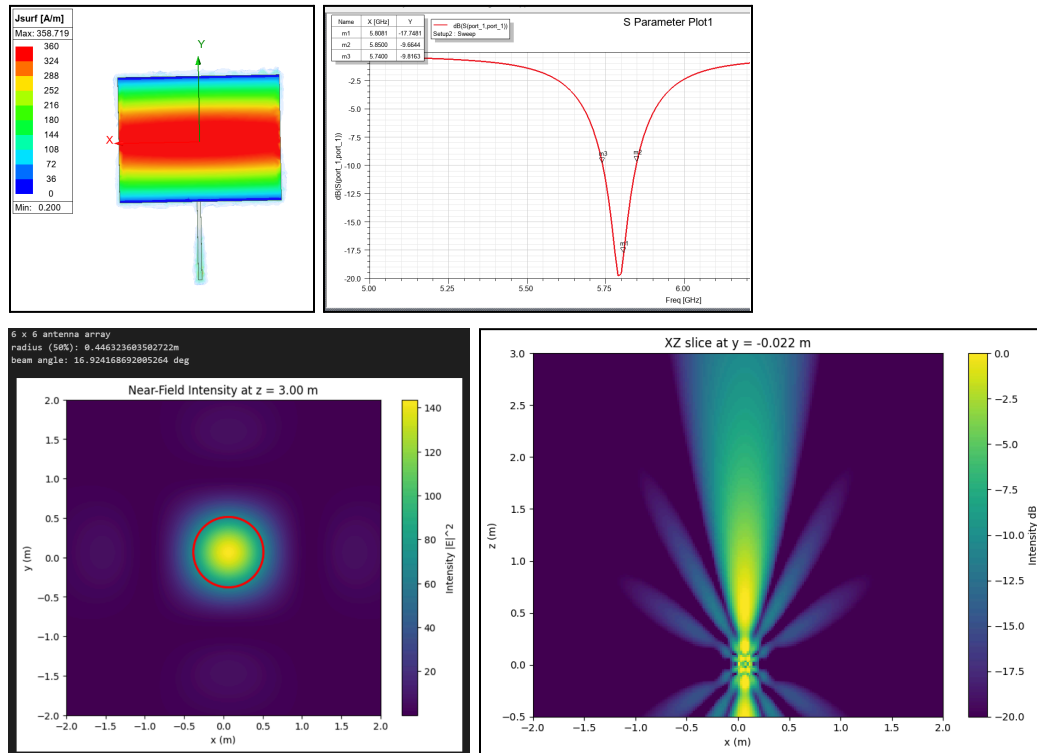
Core blocks include:

- RF signal generation
- Power amplification and attenuation
- RF splitting network
- Phase shifting network
- Impedance matching and length tuning
- Patch antenna design & array layout



3. Design Process

The design process involved lots of simulation. I used **Ansys HFSS** to optimize antenna shape, tuning, and impedance matching. I used python to simulate the array's beamforming characteristics and beam steering.

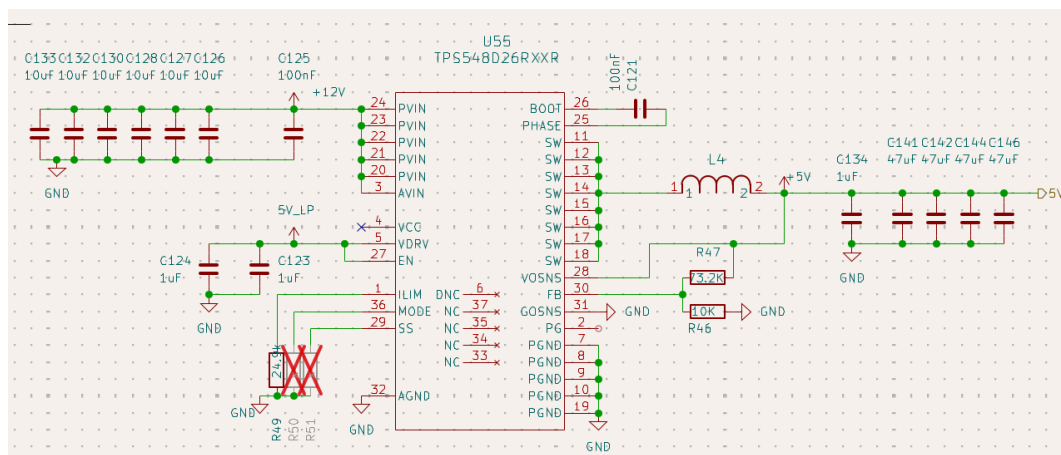


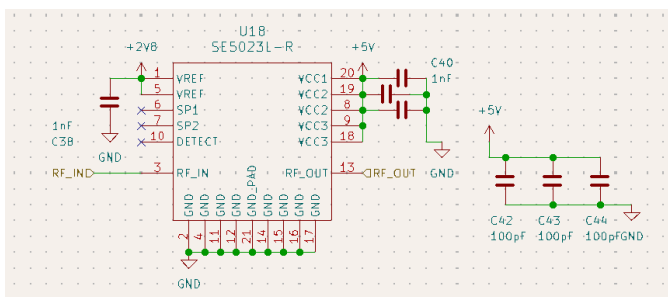
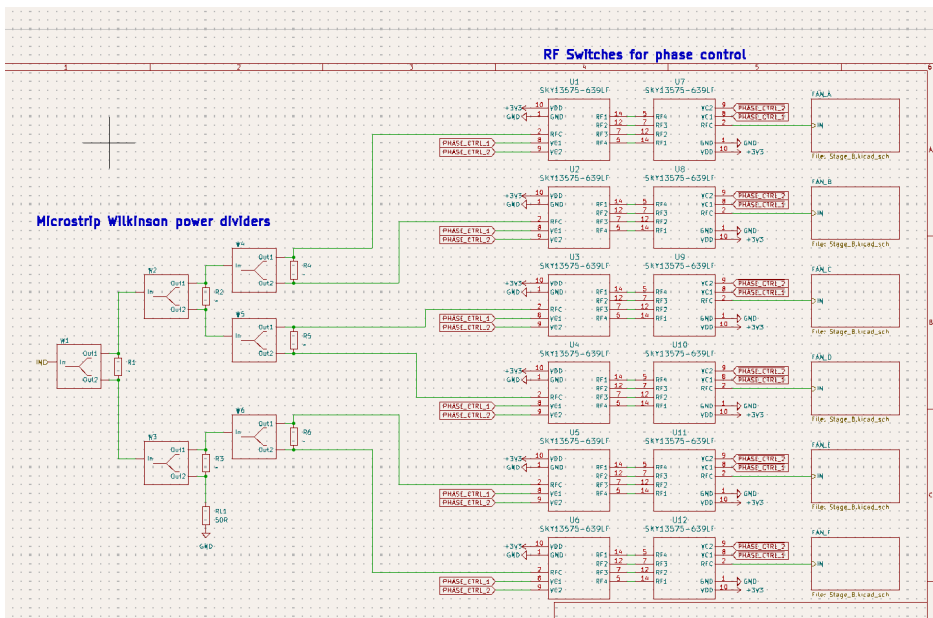
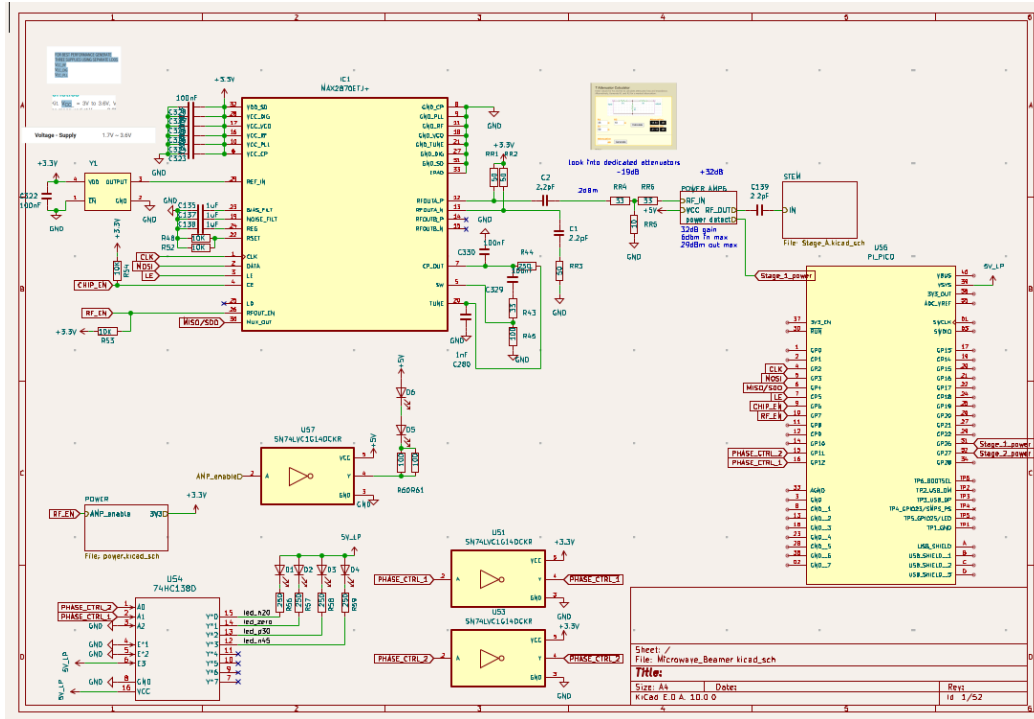
Simulating the beam from my array using python reveals a 3dB beam dispersion of 17 degrees.

3.1. Schematic design

Below I show some of the core blocks in the schematic design.

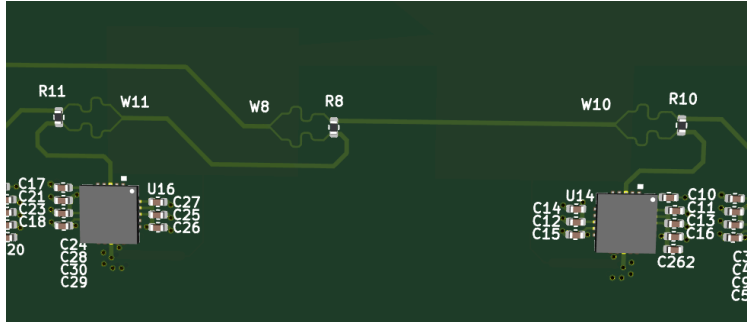
The full schematic can be seen [here](#)



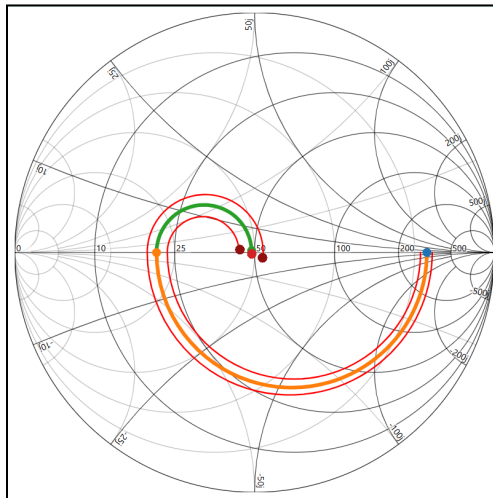
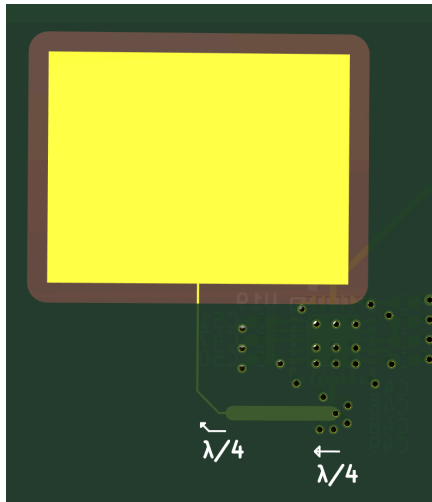


4. Microstrip Impedance Matching

I use **Wilkinson power dividers** to maintain matched impedance in the signal lines as well as provide isolation between the outputs.



Quarter wavelength transformers are used to match the ~ 300 ohm patch antennas to the 50 ohm amplifiers. QWT was chosen over stub or lumped element matching due to its robustness: 20% inaccuracies in either the length or line impedance still lands with a manageable 10dB return loss.



Phase control is achieved using SP4T switches to select the appropriate delay path for the chosen beam steering angle. The chosen steering angles in degrees are: 45, 20, 0, -30. This provides 75 degrees of steerability, and 4 unique angles.



5. Technical Challenges

- Sourcing RF components that operate reliably at the target microwave frequency
- Managing PCB dielectric tolerances affecting antenna tuning & electrical length
- Maintaining consistent phase across RF paths
- Thermal considerations under higher power levels
- Avoiding unintended radiation or coupling between array elements

Most resistors and capacitors do not provide documentation of performance at high frequencies - especially as high as 5.8GHz. Below shows impedance vs frequency charts for resistors chosen for the T attenuator, and capacitors chosen for DC blockers. Murata's "sim surf" website was very handy for choosing appropriate capacitors.

